



### New Covid-19 test

Scientists developed an experimental test for COVID-19 which uses plasmonic gold nanoparticles. <https://digit.in/jun20-34>



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merchants who traded both with the Austronesians and the Egyptians may have been responsible for the movement of these ancient objects half way across the world.

Around 300 BCE, Chandragupta Maurya established a navy, which was used to establish contact with Greece, Syria and Egypt. Indigo, silk, cotton and spices were the main items traded by the Indians. There was also an Indian presence in the port of Alexandria. Chandragupta Maurya's Prime Minister, Kautilya, better known as Chanakya, dedicated an entire chapter to shipping in his treatise on statecraft, Arthashastra.

Between the second century BCE and till as recently as 1500 CE, there was a Maritime Silk Road, which was a set of connected sea routes. The Austronesian, Indian, Greek, Roman, Arab and Persian traders and sailors were the primary components of the network, even though the name is associated with China. The network grew out of early trading voyages by the Austronesian people to India and Sri Lanka.

## CHINA

The Chinese began building trading ships during the regime of the Song Dynasty, around 800 CE. Around this time itself, the Chinese had many innovations far ahead of the Europeans. The sizes of the ships were comparable to the ones used by Columbus 800 years later. There were Chinese ships that used three to four masts, watertight compartments to prevent flooding in case the ship was rammed into, water tanks to hold fresh catch and a sternpost rudder that could be raised and lowered to allow the ship to navigate in shallow waters. All these innovations were a thousand years before they appeared in Europe. The Chinese learned what they could from the previous seafarers, including lateen sails that allowed them to sail against the direction of the winds. The design was borrowed from Arab sailors, who themselves picked up the trick from the Austronesians.

Within a few centuries, the Chinese had established trading posts in the Javanese islands, Sri Lanka and along the eastern coast of India. After the warrior prince Zhu Di usurped the throne in 1402, he appointed his right hand man (actually a eunuch) named Zheng. He as the commander of a treasure fleet. Zhu Di replaced the confucian philosophies of the previous rulers, which did not believe in trade too much. Zheng He's fleet was a large armada that made seven voyages to India and even beyond to Africa, "discovering" the Indian ocean before the European Imperialists. There were 208 vessels in the fleet, with 62 treasure ships and 27,800

of the Song dynasty was used as a navigational aid. Later on, he sailed to Bengal and into the Persian Gulf. In Bengal, the Chinese traders came across a giraffe that had been gifted to the ruler by African traders, and convinced the ruler to part with the animal, which was taken to the court in Nanjing in 1415. A follow up expedition was launched to Africa, where Zheng retrieved exotic treasures such as spices, ivory, wood, medicines and pearls. Zheng He also brought back some familiar friends to the giraffe, including lions, leopards, ostriches, zebras and oryxes. According to the Chinese, the giraffe was the famed unicorn, the ostrich was a camel



A replica of Zheng He's treasure ship

crewmen. It was a massive undertaking. The treasure ships were reported to be so enormous that scientists simply did not believe them. The dimensions were in excess of the 400 feet in length and 150 feet in width, which would make the treasure ships larger than the biggest wooden galleons built in Europe. The larger ships had many structural problems even at the height of European shipbuilding. However, a rudder was discovered in Nanjing, which measured 30 feet in length, which would make a treasure ship around 500 feet in length.

In his maiden voyage, Zheng He sailed to Vietnam and Calicut in India. A compass invented in the time

bird, and the zebras and oryxes were heavenly horses and deer. They saw all of it as good omens, but they were, in fact, not. Zheng He died towards the end of his seventh voyage, and the Confucians took over again. They destroyed all the larger ships, prevented boats with masted sails from setting out from China, and began a regime of isolation that lasted for centuries. Even the records of Zheng He's expeditions and routes were destroyed. Wherever he went, Zheng He established good relations in Thailand, India, Arabia, Lakshadweep, Maldives and Africa. He ferried envoys from these distant lands to the Imperial court and then returned them as

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